PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by Act No. 41 of 2013
Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

CIVIC-VOICE An AI Assistant for Multi-Channel Citizen Feedback Analysis in Public Services

A PROJECT REPORT

Submitted by

INCHARA K R - 20231CAI0117

PEDDI REDDY TEJASWINI - 20231CAI0120

SAHANA HARAGAPURE - 20231CAI0115

Under the guidance of,

JAGADEVI BAKKA

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING
(Artificial Intelligence and Machine Learning)

PRESIDENCY UNIVERSITY

BENGALURU

DECEMBER 2026

PRESIDENCY SCHOOL OF ARTIFICIAL INTELLIGENCE & ADVANCED COMPUTING

BONAFIDE CERTIFICATE

Certified that this report " CIVIC-VOICE An AI Assistant for Multi-Channel Citizen Feedback Analysis in Public Services" is a bonafidework of "INCHARA KR (20231CAI0117), PEDDI REDDY TEJASWINI (20231ICAI0120), SAHANA HARAGAPURE (20231CAI0115)", who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING) during 2026-27.

Ms Jagadevi Bakka
Assistant Professor
Project Guide PSAIAC
Presidency University

Ms Suma Gowda
Assistant Professor
Program Project Coordinator
PSAIAC
Presidency University

Ms Rakheeba Taseen
Assistant Professor
School Project Coordinator
PSAIAC
Presidency University

Dr Zafar Ali Khan
Professor & Head of the Department
PSAIAC
Presidency University

Dr. Shakkeera L
Dean
PSAIAC
Presidency University

Sl. No	Name of the Examiner	Signature	Date
1	Dr.Mohammed Mujeerulla		
2	Dr.Pakruddin B		

PRESIDENCY SCHOOL OF ARTIFICIAL INTELLIGENCE & ADVANCED COMPUTING

DECLARATION

We the students of final year B.Tech in COMPUTER SCIENCE AND ENGINEERING (Artificial Intelligence and Machine Learning) at Presidency University, Bengaluru, INCHARA KR, PEDDI REDDY TEJASWINI , SAHANA HARAGAPURE, hereby declare that the project work titled "CIVIC-VOICE An AI Assistant for Multi-Channel Citizen Feedback Analysis in Public Services" has been independently carried out by us and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING) during the academic year of 2026-27. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

INCHARA KR	USN: 20231CAI0117	Signature 1
PEDDI REDDY TEJASWINI	USN: 20231CAI0120	Signature 2
SAHANA HARAGAPURE	USN: 20231CAI0115	Signature 3

PLACE: BENGALURU
DATE:

ACKNOWLEDGEMENT

For completing this project work, we have received the support and the guidance from many people whom I would like to mention with deep sense of gratitude and indebtedness. We extend our gratitude to our beloved **Chancellor, Pro-Vice Chancellor, and Registrar** for their support and encouragement in completion of the project.

I would like to sincerely thank my internal guide **Ms Jagadevi Bakka, Assistant Professor**, Presidency School of Artificial Intelligence & Advanced Computing, Presidency University, for his/her moral support, motivation, timely guidance and encouragement provided to us during the period of our project work.

I am also thankful to **Dr. Zafar Ali Khan**, **Professor**, **Head of the Department**, Presidency School of Artificial Intelligence & Advanced Computing, Presidency University, for his mentorship and encouragemen

We express our cordial thanks to **Dr. Shakkeera L**, **Dean**, Presidency School of Artificial Intelligence & Advanced Computing, Presidency University for providing the required facilities and intellectually stimulating environment that aided in the completion of my project work.

We are grateful to **Ms. Rakheeba Taseen, PSAIAC School Project Coordinator, PSAIAC Program Project Coordinator**, for facilitating problem statements, coordinating reviews, monitoring progress, and providing their valuable support and guidance.

We are also grateful to Teaching and Non-Teaching staff of Presidency School of Artificial Intelligence & Advanced Computing and also staff from other departments who have extended their valuable help and cooperation.

ABSTRACT (16)

CIVIVOICE is an AI-driven multi-channel citizen feedback analysis system designed to collect, preprocess, analyze, and interpret citizen feedback from multiple digital and public-service sources. Citizens submit complaints, grievances, opinions, and suggestions through different platforms, resulting in large volumes of unstructured textual data. Manually analyzing such data is time-consuming and makes it difficult for organizations to identify recurring issues, prioritize urgent complaints, and understand citizen concerns effectively. CIVIVOICE addresses this challenge by developing a centralized Natural Language Processing (NLP) pipeline for automated feedback analysis.

The system integrates feedback from multiple sources, including CPGRAMS grievances, Bengaluru local complaints, surveys, Twitter, and email datasets. The collected data undergoes data cleaning, normalization, duplicate analysis, merging, validation, and text preprocessing to produce a unified master dataset. The current master dataset contains approximately 205,000 feedback records. NLP preprocessing is applied to clean and normalize textual information while preserving the important meaning of citizen feedback. Further analysis includes exploratory text analysis and TF-IDF-based identification of important terms and patterns. The system is designed to extend this pipeline with sentiment analysis, complaint categorization, urgency detection, keyword extraction, and other NLP-based analytical techniques.

A MySQL database is proposed for storing the processed feedback in a structured and scalable manner. Python will be used for data processing and NLP model development, while a Streamlit-based dashboard will provide an interactive interface for visualizing the results. The dashboard will enable users to examine feedback trends, complaint categories, urgency levels, sources, locations, and other relevant insights.

By combining multi-source data integration, NLP, machine learning, database management, and interactive visualization, CIVIVOICE aims to transform large-scale unstructured citizen feedback into meaningful and actionable information. The system can assist organizations and public authorities in identifying major citizen concerns, understanding emerging patterns, and supporting data-driven decision-making for improved public service delivery.Public-service organizations receive large volumes of citizen feedback through multiple communication channels, including email, social media platforms, grievance systems, surveys, and direct citizen submissions. Such feedback contains valuable information about citizens' experiences, complaints, requests, and expectations regarding public services. However, the heterogeneous and largely unstructured nature of this information makes it difficult for authorities to manually process, categorize, prioritize, and identify recurring problems and emerging issues. To address this challenge, this project proposes **CivicVoice**, an AI-powered multi-channel citizen feedback analysis and public-service intelligence platform designed to transform unstructured citizen feedback into structured and actionable insights.

The proposed platform therefore combines **multi-channel data integration**, NLP-based analysis, centralized database management, visualization, and AI-assisted querying within a single framework. The project aims to improve the accessibility and usefulness of citizen feedback by converting large volumes of textual information into structured insights that can support public-service monitoring and decision-making. The system will be evaluated using appropriate NLP, classification, analytical, and system-performance metrics, with detailed experimental results to be incorporated following implementation and testing.

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1.Introduction

1.1 Background

Citizen feedback plays an important role in understanding the quality, accessibility, and effectiveness of public services. Citizens communicate their experiences, complaints, suggestions, and concerns through several channels, including government grievance portals, local complaint systems, surveys, email, and social media platforms. These sources contain valuable information about recurring public-service problems, but the information is often distributed across different systems and stored in different formats. As a result, analysing citizen feedback manually can be time-consuming and may make it difficult to identify common issues and emerging trends.

CIVIC-VOICE is proposed as a multi-channel citizen feedback analysis system that brings information from different sources into a unified analytical framework. The project combines data engineering and Natural Language Processing (NLP) techniques to clean, normalize, validate, and analyse citizen feedback. The current project integrates feedback from five major sources: CPGRAMS, Bengaluru local complaints, survey responses, Twitter data, and email feedback. These sources together form a master dataset containing approximately 205,168 records.

The project focuses on creating a reliable foundation for analysing citizen feedback. Data preprocessing includes normalization, duplicate analysis, merging, validation, and text cleaning. The processed dataset can subsequently support activities such as text analysis, keyword extraction, categorization, urgency analysis, and visualization. Through this approach, CIVIC-VOICE aims to make large-scale citizen feedback easier to organize, interpret, and use for public-service analysis.

1.2 Problem Statement

Public-service organizations receive large volumes of citizen feedback through multiple communication channels. Although this feedback contains important information about service quality and public concerns, it is often difficult to analyse because the information is distributed across different platforms and stored using different structures. Government grievance portals, local complaint databases, surveys, emails, and social media platforms may use different fields, formats, timestamps, categories, and levels of detail. This creates difficulties when attempting to combine the information into a single analytical dataset.

Another important challenge is the presence of data-quality issues. Citizen feedback may contain duplicate records, missing values, inconsistent text formats, incomplete locations, and unavailable urgency information. Unstructured textual feedback also requires preprocessing before it can be effectively analysed using Natural Language Processing techniques. Manual examination of thousands of complaints is inefficient and may not provide a consistent understanding of recurring issues.

CIVIC-VOICE addresses this problem by developing a structured pipeline for collecting, normalizing, cleaning, validating, and integrating citizen feedback from multiple sources. The project currently combines five sources into a master dataset of approximately 205,168 records. The system applies data preprocessing and NLP techniques to transform heterogeneous citizen feedback into a consistent format suitable for further analysis. This provides a foundation for identifying patterns, extracting important terms, studying urgency, and supporting data-driven public-service analysis.

1.3 Existing System

Existing citizen-feedback systems generally operate around individual platforms or specific channels through which citizens submit their complaints and suggestions. Government grievance portals, for example, primarily manage complaints submitted through their own platforms, while local complaint systems may focus on issues reported within a particular geographic area. Similarly, survey systems collect structured responses according to predefined questions, whereas social media and email contain largely unstructured textual feedback. Although these systems are useful for collecting information, they commonly operate independently of one another.

The separation of feedback across different platforms creates difficulties when organizations need to obtain a broader understanding of citizen concerns. Data from different sources may have different schemas, field names, formats, and levels of completeness. Some datasets may also contain duplicate records or missing information. In addition, textual feedback requires additional processing before meaningful patterns can be extracted.

Traditional analysis may depend heavily on manual inspection, simple filtering, or platform-specific reporting tools. Such approaches can become inefficient when the number of records increases significantly. They may also make cross-source comparison difficult.

CIVIC-VOICE is designed to address these limitations by providing a unified framework for multi-source citizen feedback analysis. Rather than treating each feedback channel independently, the project combines different datasets, applies common preprocessing procedures, validates the resulting data, and prepares the information for NLP-based analysis and visualization.

1.4 Proposed CivicVoice Approach

The proposed CIVIC-VOICE approach focuses on integrating citizen feedback from multiple sources into a unified and structured analytical pipeline. The system begins by collecting feedback from different channels, including CPGRAMS, Bengaluru local complaints, survey responses, Twitter, and email. Since each source has a different structure, source-specific normalization is performed to make the information consistent and suitable for further processing.

After ingestion and normalization, the datasets undergo data-quality analysis. This includes identifying duplicate records, examining missing values, validating important fields, and checking the consistency of the merged data. The cleaned datasets are then combined to create a master dataset containing approximately 205,168 records. This master dataset provides a common foundation for subsequent analysis.

The next stage focuses on text preprocessing. Citizen feedback is transformed into a cleaner textual representation by applying appropriate normalization and cleaning operations. The processed text can then be used for exploratory analysis and Natural Language Processing tasks. Techniques such as term-frequency analysis and TF-IDF can be used to identify important words and patterns within the feedback.

The overall approach therefore connects data engineering with NLP-based analysis. Instead of analysing each feedback source separately, CIVIC-VOICE creates a common data pipeline that allows information from different channels to be studied together. The architecture also provides a foundation for future database integration, dashboard development, categorization, sentiment analysis, and urgency prediction.

1.5 Objectives

The primary objective of CIVIC-VOICE is to develop a structured framework for collecting, integrating, preprocessing, and analysing citizen feedback obtained from multiple channels. The project aims to transform heterogeneous feedback data into a consistent dataset that can support further analysis and public-service decision-making.

One of the main objectives is to integrate feedback from different sources, including CPGRAMS, Bengaluru local complaints, survey responses, Twitter, and email. Since these sources contain different structures and formats, the project aims to apply source-specific normalization and common validation procedures. Another objective is to identify and address data-quality issues such as duplicate records, missing values, inconsistent fields, and variations in textual content.

The project also aims to develop an effective text-preprocessing pipeline for citizen feedback. This includes cleaning and normalizing textual information so that it can be used for Natural Language Processing. The processed dataset can then support exploratory analysis, frequency analysis, TF-IDF-based feature extraction, and identification of meaningful patterns in citizen feedback.

A further objective is to create a reliable master dataset that can serve as the foundation for future analytical components. The project also provides a basis for developing visualization and dashboard capabilities. In later stages, the system can be extended with automated feedback categorization, sentiment analysis, urgency prediction, topic analysis, database integration, and an interactive dashboard for exploring citizen concerns.

1.6 Sustainable Development Goals (SDGs)

CIVIC-VOICE is related to sustainable development because effective public services depend on understanding the experiences and concerns of citizens. A structured citizen-feedback analysis system can support organizations in identifying recurring service issues and understanding patterns within large volumes of public feedback. By organizing information from multiple communication channels, the project provides a foundation for more systematic analysis of citizen concerns.

The project has particular relevance to Sustainable Development Goal 11, which focuses on making cities and human settlements inclusive, safe, resilient, and sustainable. Feedback from citizens can provide information about issues affecting local services and urban environments. By bringing together complaints and feedback from multiple sources, CIVIC-VOICE can support the analysis of such concerns at scale.

The project is also connected with Sustainable Development Goal 16, which promotes peaceful and inclusive societies, access to justice, and effective, accountable, and inclusive institutions. A structured mechanism for analysing citizen feedback can contribute to more informed understanding of public concerns and support evidence-based approaches to improving service delivery.

CIVIC-VOICE does not directly achieve these Sustainable Development Goals; rather, its contribution is through the development of a technological framework that can support citizen participation, data-driven analysis, and public-service monitoring. The integration of multi-source feedback, data validation, NLP preprocessing, and analytical techniques provides a foundation for future systems that can help organizations better understand and respond to citizen needs.

1.7 Report Organization

This report presents the development of the CIVIC-VOICE project from its background and problem definition through its methodology, implementation, evaluation, and future scope. The report is organized into multiple chapters to provide a structured description of the system and the work completed during the project.

Chapter 1 introduces CIVIC-VOICE by describing the background of multi-channel citizen feedback analysis, the problem being addressed, existing approaches, the proposed system, project objectives, relevant Sustainable Development Goals, and the organization of the report.

Chapter 2 presents the literature review and discusses existing research and systems related to citizen feedback analysis, Natural Language Processing, multi-source data integration, and public-service feedback management. It also identifies the research gap addressed by CIVIC-VOICE.

Chapter 3 describes the methodology followed in the project, including data collection, source-wise normalization, data cleaning, duplicate analysis, data merging, validation, and NLP preprocessing.

Chapter 4 discusses project planning and management activities.

Chapter 5 focuses on system analysis and design, including requirements, architecture, data flow, and database design.

Chapter 6 describes the implementation of the major project components and processing scripts.

Chapter 7 presents the results and evaluation of the datasets, preprocessing pipeline, and analytical methods.

Chapter 8 discusses social, legal, ethical, sustainability, privacy, and safety considerations. Finally,

Chapter 9 concludes the report by summarizing the work, discussing limitations, and presenting possible future enhancements to CIVIC-VOICE.

2.LITERATURE REVIEW

2.1 Citizen Feedback Analysis

Citizen feedback analysis has become an important area for improving the understanding of public-service requirements and identifying recurring issues reported by citizens. Citizen feedback can include complaints, suggestions, opinions, service requests, and general observations about public services. Such information can provide organizations with valuable insights into areas where services may require improvement. However, the increasing volume of citizen-generated information makes manual analysis difficult and time-consuming.

Traditional approaches to citizen feedback analysis often focus on individual complaint-management systems. These systems generally collect and organize feedback within a particular platform but may not provide a unified view across multiple communication channels. As citizens increasingly use government portals, surveys, email, local complaint systems, and social media, the need for multi-source analysis becomes more important.

CIVIC-VOICE addresses this requirement by combining citizen feedback obtained from multiple sources into a common analytical dataset. The project considers feedback from CPGRAMS, Bengaluru local complaints, surveys, Twitter, and email. These sources differ in their structure and content, requiring preprocessing before analysis.

The literature and existing approaches demonstrate the importance of structured citizen-feedback management, while CIVIC-VOICE extends this concept by focusing on multi-source data integration, data-quality processing, text preprocessing, and NLP-based analysis. This provides a foundation for studying large volumes of citizen feedback in a consistent manner.

2.2 NLP-Based Feedback Analysis

Natural Language Processing (NLP) provides techniques for processing and analysing human language in textual data. Since a large portion of citizen feedback is provided as free-form text, NLP is particularly useful for extracting meaningful information from complaints, suggestions, and opinions. Without text processing, it can be difficult to systematically identify frequently discussed issues across thousands of records.

Common NLP operations include text cleaning, normalization, tokenization, stop-word handling, stemming or lemmatization, and feature extraction. These operations convert raw textual information into a representation that can be processed by analytical and machine-learning methods. Feature extraction techniques such as Term Frequency-Inverse Document Frequency (TF-IDF) can be used to identify terms that are informative within a collection of documents.

In CIVIC-VOICE, NLP is applied after the multi-source data has been collected, normalized, merged, and validated. The project includes a text-preprocessing stage that produces cleaned textual data for further analysis. Exploratory text analysis can then be performed to identify frequently occurring terms and patterns.

NLP also provides a foundation for future components of CIVIC-VOICE, including automated complaint categorization, sentiment analysis, urgency prediction, and topic identification. However, these advanced models should be considered separate stages that require appropriate labelled data and evaluation. The current work therefore establishes the preprocessing and feature-extraction foundation required for subsequent NLP-based analysis.

2.3 Multi-Channel Data Integration

Multi-channel data integration refers to the process of combining information collected from different sources into a unified data environment. In the context of citizen feedback, multiple channels may contain overlapping or complementary information. Government grievance portals may provide structured complaint information, while social media, email, and survey responses may contain more informal or detailed descriptions of citizen experiences.

A major challenge in multi-channel integration is that different sources generally use different schemas, field names, formats, and data representations. One source may contain detailed timestamps and locations, while another may have incomplete information. Text fields may also vary considerably in length and structure. Therefore, simply combining datasets without preprocessing can result in inconsistent or unreliable analytical results.

CIVIC-VOICE addresses this issue through a structured data-integration pipeline. Each source is first processed using source-specific normalization procedures. Duplicate analysis and validation are then performed before the datasets are merged into a common master dataset. The resulting dataset provides a consistent foundation for NLP and exploratory analysis.

The project currently integrates five sources: CPGRAMS, Bengaluru local complaints, survey data, Twitter data, and email feedback. The combined master dataset contains approximately 205,168 records. This multi-channel approach allows the project to move beyond source-specific analysis and provides a common framework for examining citizen feedback from different communication environments.

2.4 Existing Systems

Existing citizen-feedback systems include government grievance portals, local complaint-management platforms, survey applications, and social-media-based feedback mechanisms. These systems provide channels through which citizens can report problems, submit complaints, provide suggestions, or express their experiences with public services. Each system serves an important role in collecting feedback, but the information is often maintained separately.

Government grievance systems generally provide structured mechanisms for registering and tracking complaints. Local complaint systems may focus on specific geographic areas or service categories. Surveys provide controlled and structured responses, while email and social media can provide more open-ended feedback. The different characteristics of these systems make direct comparison and combined analysis difficult.

A common limitation is the lack of a unified analytical layer across multiple feedback channels. Data may remain within separate databases or files, making it difficult to identify patterns that appear across different sources. Differences in schemas, missing values, duplicate records, and textual formats further increase the complexity of cross-source analysis.

CIVIC-VOICE is designed to provide such a common analytical layer. Rather than replacing existing feedback-collection systems, the project focuses on integrating their outputs. Through normalization, validation, duplicate analysis, merging, and NLP preprocessing, feedback from different systems can be transformed into a common dataset. This creates a foundation for further analysis and visualization while retaining the contribution of each individual source.

2.5 Research Gap

The review of citizen-feedback analysis approaches highlights a gap between feedback collection and unified multi-source analysis. Many existing systems concentrate on collecting and managing feedback within individual platforms. Although such systems can effectively handle their respective data, they may not provide a common framework for integrating information from government portals, local complaints, surveys, email, and social media.

Another gap exists in the treatment of unstructured feedback. Citizen comments often contain natural language that cannot be directly analysed using simple structured-data techniques. Data cleaning, normalization, duplicate detection, and NLP preprocessing are therefore necessary before meaningful patterns can be extracted. A system that combines these stages can provide a more consistent foundation for large-scale citizen-feedback analysis.

CIVIC-VOICE addresses this gap by developing an integrated pipeline for multi-source feedback processing. The project combines five feedback sources and creates a master dataset of approximately 205,168 records. The workflow includes source-wise normalization, duplicate analysis, data merging, validation, and text preprocessing.

The project also establishes a foundation for future NLP-based analytical capabilities. Rather than treating sentiment analysis, categorization, or urgency prediction as already completed features, these are positioned as future analytical extensions requiring suitable models and evaluation. Thus, the main contribution of the current work is the development of a structured, validated, multi-source dataset and processing pipeline that can support subsequent citizen-feedback intelligence.

3. METHODOLOGY

3.1 Overall System Methodology

The CIVIC-VOICE methodology follows a sequential data-processing pipeline designed to transform heterogeneous citizen feedback into a structured dataset suitable for analysis. The methodology begins with data collection from multiple sources and continues through ingestion, cleaning, normalization, duplicate analysis, merging, validation, and NLP preprocessing.

The first stage involves obtaining citizen feedback from five sources: CPGRAMS, Bengaluru local complaints, survey responses, Twitter, and email. Since these sources have different structures, each dataset is processed separately before integration. Source-specific normalization ensures that important fields are converted into a consistent representation.

After normalization, the datasets undergo data-quality analysis. Duplicate records are examined to identify repeated entries, while missing values and other inconsistencies are evaluated. The cleaned datasets are subsequently merged into a unified master dataset. The current master dataset contains approximately 205,168 records and seven columns.

The next stage involves NLP preprocessing of citizen feedback text. The preprocessing workflow generates cleaned text while preserving the overall record count. Validation is then performed to confirm that the processed dataset remains consistent with the source data.

Finally, exploratory text analysis and feature-extraction techniques such as TF-IDF can be applied. The methodology therefore creates a complete foundation connecting data engineering with NLP analysis. Future stages can extend this pipeline with classification, sentiment analysis, urgency prediction, database integration, and interactive visualization.

3.2 Data Collection

Data collection is the first major stage of the CIVIC-VOICE methodology. The objective of this stage is to gather citizen feedback from different channels so that the project can analyse public-service concerns from multiple perspectives. Instead of depending on a single feedback mechanism, the project combines information from several sources.

The current CivicVoice dataset includes CPGRAMS data, Bengaluru local complaint data, survey responses, Twitter data, and email feedback. These sources represent different types of citizen interaction. Government grievance data can contain formal complaints, local complaint data represents geographically focused feedback, surveys provide structured responses, while Twitter and email may contain more informal or free-form textual feedback.

The collected datasets vary in size, schema, format, and completeness. Therefore, the raw data cannot be directly merged. Each dataset first undergoes inspection to identify available fields, data types, textual columns, timestamps, locations, and other relevant attributes.

The collected datasets form the foundation of the subsequent processing pipeline. The project currently contains approximately 205,168 records in the final master dataset after integration. During collection and preparation, attention is given to preserving source information so that records can later be traced back to their respective channels.

The data-collection stage therefore establishes the multi-source nature of CIVIC-VOICE and provides the raw material required for subsequent cleaning, normalization, NLP processing, and analysis.

3.3 Data Ingestion

Data ingestion refers to the process of bringing the collected datasets into the CIVIC-VOICE processing environment. Since the project uses multiple data sources, ingestion is designed to accommodate differences in file structure and source-specific characteristics.

Each dataset is first loaded into the processing environment and examined for its available fields and data types. The ingestion process preserves the original information required for later processing while preparing the data for source-specific normalization. This approach is important because directly applying one common transformation to all sources may result in the loss or incorrect interpretation of source-specific information.

The project uses separate processing scripts for different datasets. Examples include scripts for normalizing CPGRAMS data, local complaints, survey data, and merging feedback. This modular approach makes the data pipeline easier to maintain and allows source-specific transformations to be performed independently.

During ingestion, important attributes such as feedback text, timestamps, locations, source identifiers, and urgency-related information are considered. The availability of these attributes differs between sources, which is reflected in subsequent data-quality analysis.

The ingestion stage ultimately prepares the individual datasets for normalization and integration. It also establishes a clear separation between raw source data and processed data. This separation is useful for reproducibility because transformations can be reviewed and repeated without altering the original source information.

3.4 Data Cleaning

Data cleaning is performed to improve the consistency and reliability of the datasets before they are merged and analysed. Citizen feedback collected from different channels may contain incomplete fields, inconsistent formatting, unwanted characters, duplicated records, or variations in textual representation. Cleaning is therefore an important step in the CIVIC-VOICE pipeline.

The cleaning process begins with an examination of the individual datasets. Fields are checked for missing or invalid values, and textual information is inspected for formatting inconsistencies. Source-specific cleaning is applied where required so that information from each channel can be retained without introducing inappropriate transformations.

Data-quality validation of the final master dataset identified several missing-value patterns. For example, the master dataset contains records with unavailable timestamps, locations, and urgency information. These values are not automatically treated as invalid because their absence may originate from the source dataset itself. Instead, the missing information is documented and considered during analysis.

Textual data also requires cleaning before NLP processing. The project applies a dedicated text-preprocessing workflow to transform raw feedback into cleaner text. This allows subsequent feature extraction and exploratory analysis to operate on more consistent input.

The objective of cleaning is therefore not to remove information unnecessarily, but to identify and handle inconsistencies systematically. The cleaned data provides a more reliable foundation for duplicate analysis, merging, NLP preprocessing, and statistical analysis.

3.5 Data Normalization

Data normalization is the process of converting information from different sources into a consistent structure and representation. This stage is particularly important in CIVIC-VOICE because the five input datasets were not originally designed using one common schema.

Separate normalization scripts are used for different sources. For example, CPGRAMS data and Bengaluru local complaint data require different transformations because their original structures and field representations differ. Survey data and other feedback sources similarly require source-specific processing.

Normalization includes activities such as standardizing field representations, preparing textual fields, handling source-specific attributes, and creating common fields required for later merging. The objective is to ensure that equivalent information from different datasets can be represented consistently in the master dataset.

An important aspect of normalization is preserving the original source identity. The source information allows analysts to understand where individual feedback records originated and makes it possible to perform source-wise analysis later.

Normalization also prepares the data for validation. Once the individual datasets follow a compatible structure, they can be checked for completeness and consistency before being combined. This reduces the possibility of schema conflicts during merging.

The CIVIC-VOICE normalization stage therefore acts as a bridge between raw source data and the unified master dataset. It makes heterogeneous datasets suitable for common downstream operations while retaining important source-specific information required for interpretation and analysis.

3.6 Duplicate Analysis

Duplicate analysis is performed to identify repeated records within the citizen-feedback datasets. Duplicate records can occur when information is collected from the same source more than once, when records are exported repeatedly, or when similar information is represented through multiple entries. If duplicates are not examined, they may distort frequency-based analysis and influence the interpretation of citizen concerns.

CIVIC-VOICE includes a dedicated duplicate-analysis stage before final dataset preparation. The project uses an analysis script to examine duplicate patterns within the datasets. The results of this analysis are considered when preparing the merged data.

Duplicate analysis is particularly important in a multi-source system because records from different channels may also contain similar or overlapping information. However, identical text does not automatically mean that two records represent the same citizen interaction. Therefore, duplicate handling must consider the available identifiers and contextual fields rather than relying only on textual similarity.

The purpose of the duplicate-analysis stage is to understand the quality of the input datasets and prevent unnecessary repetition from affecting later analysis. The findings are documented as part of the overall data-quality assessment.

By including duplicate analysis as an explicit methodological step, CIVIC-VOICE provides greater transparency regarding how the master dataset was constructed. This also improves the reliability of subsequent frequency analysis, TF-IDF processing, and any future machine-learning models trained using the dataset.

3.7 Data Merging

Data merging is the stage in which the individually normalized datasets are combined into a unified CivicVoice dataset. Since the project uses five different feedback sources, merging provides the central mechanism through which information from different channels becomes available for common analysis.

Before merging, the datasets are processed independently to resolve source-specific structural differences. Fields that are common across sources are mapped into a consistent representation, while source-specific information is retained where appropriate. A source identifier is particularly useful because it allows the combined dataset to be separated and analysed according to its original channel.

The merging process combines the processed CPGRAMS, Bengaluru local complaint, survey, Twitter, and email datasets. After integration, the project produces a master dataset containing approximately 205,168 records across seven columns.

The merged dataset is subsequently validated to ensure that the integration process has not unexpectedly changed the record structure. Record counts, missing values, textual fields, and other important attributes are examined.

The merging stage is significant because it transforms the project from a collection of independent datasets into a common analytical resource. Instead of developing separate analysis pipelines for each feedback source, subsequent NLP and exploratory analysis can operate on the unified dataset. At the same time, source information can be retained for comparative analysis. This provides the foundation for future dashboards and analytical models within CIVIC-VOICE.

3.8 Master Dataset Creation

The master dataset represents the central data resource created by the CIVIC-VOICE project. It is generated by combining the normalized and processed datasets from the five selected feedback sources. The final master dataset contains approximately 205,168 records and seven columns.

The creation of the master dataset involves several stages rather than simply concatenating raw files. First, the individual sources are collected and inspected. Source-specific normalization is then performed, followed by cleaning and duplicate analysis. The resulting datasets are merged into a common structure. Validation is performed after merging to identify missing values and confirm the consistency of the resulting data.

The master dataset provides a unified representation of citizen feedback. It contains information originating from government grievance data, local complaints, surveys, Twitter, and email. This allows the same preprocessing and analytical procedures to be applied across multiple channels.

The dataset also provides a foundation for identifying data-quality characteristics. Current validation indicates that some fields contain substantial missing information. For example, urgency information is unavailable for a large proportion of records because the original sources do not consistently provide this attribute. Such characteristics are documented rather than automatically interpreted as errors.

The master dataset is therefore one of the key outputs of the current project. It provides the structured foundation required for NLP preprocessing, exploratory analysis, feature extraction, and future machine-learning and dashboard components.

3.9 NLP Preprocessing

NLP preprocessing converts raw citizen-feedback text into a cleaner representation suitable for computational analysis. This stage is important because citizen feedback can contain inconsistent capitalization, punctuation, special characters, unnecessary formatting, and other textual variations.

CIVIC-VOICE uses a dedicated text-preprocessing workflow to process the feedback text after the master dataset has been created. The preprocessing stage produces a cleaned text representation while retaining the original record structure. This allows the original feedback to remain available while providing a standardized version for NLP analysis.

The current preprocessing results demonstrate that the overall record count is preserved. The input contained 205,168 records and the processed output also contained 205,168 records. Approximately 184,072 records had changes applied during preprocessing, demonstrating that a substantial portion of the textual data required normalization. Only one record resulted in empty cleaned text after processing.

Text preprocessing is essential before applying methods such as term-frequency analysis and TF-IDF because inconsistent textual representations can cause the same concept to appear as different features. Cleaning reduces such variation and improves the consistency of subsequent analysis.

The processed text can later support tokenization, feature extraction, classification, sentiment analysis, topic modelling, and other NLP tasks. Therefore, the current preprocessing stage establishes the linguistic foundation for future analytical extensions of CIVIC-VOICE.

3.10 NLP Validation

NLP validation is performed to ensure that text preprocessing has produced a usable dataset without unintentionally changing the overall data structure. Validation is important because preprocessing operations can potentially remove information, create empty records, or change the number of observations.

In CIVIC-VOICE, the NLP validation stage compares the original and processed datasets. One of the main checks is the number of records before and after preprocessing. The original dataset contains 205,168 records, and the processed dataset retains the same number of records. This indicates that the preprocessing pipeline did not remove records from the dataset.

The validation also examines whether cleaned text is available for the records. The current results show that only one record has empty cleaned text. Changes to the text representation are also measured. Approximately 184,072 records were modified during preprocessing, indicating that the cleaning operations affected a large proportion of the textual content.

Text-length statistics provide another useful validation measure. The original text has an average length of approximately 729 characters, while the cleaned text has an average length of approximately 716 characters. This relatively small difference indicates that preprocessing modified the textual representation without substantially reducing the overall information content.

These validation measures provide evidence that the preprocessing pipeline is functioning consistently and establish confidence in the processed dataset for subsequent text analysis.

3.11 TF-IDF / Feature Extraction

Feature extraction converts textual information into numerical representations that can be analysed computationally. One of the techniques considered within CIVIC-VOICE is Term Frequency-Inverse Document Frequency (TF-IDF). TF-IDF assigns weights to terms based on their frequency within individual documents and their distribution across the overall collection.

Term Frequency represents how often a particular term occurs in a document. In contrast, Inverse Document Frequency reduces the importance of terms that occur in many documents and increases the relative importance of terms that are more distinctive. The combined TF-IDF value therefore helps identify terms that provide useful information for distinguishing documents.

In CIVIC-VOICE, TF-IDF can be applied to the cleaned citizen-feedback text after preprocessing and validation. The resulting feature matrix can support exploratory analysis by identifying important terms within the collection. It can also provide numerical input for future machine-learning models.

Feature extraction is particularly useful for citizen feedback because raw text cannot be directly processed by many traditional machine-learning algorithms. Converting the text into numerical features creates a bridge between NLP and predictive analysis.

The TF-IDF stage also provides an opportunity to compare vocabulary patterns across different feedback sources. For example, terms that are important in government complaints may differ from those found in social-media feedback. Such comparisons can contribute to understanding differences in citizen concerns while maintaining the unified structure of the CivicVoice dataset.

4. PROJECT MANAGEMENT

4.1 Project Planning

Project planning provides a structured approach for managing the development of CIVIC-VOICE from data collection through analysis and system implementation. Because the project combines data engineering, NLP, database development, and visualization, planning is required to ensure that the different components are developed in an appropriate sequence.

The project begins with identifying the problem of fragmented citizen feedback and defining the objectives of the CivicVoice system. The next stage involves identifying suitable feedback sources and preparing the datasets for analysis. Data collection and source-specific normalization are followed by cleaning, duplicate analysis, merging, and validation.

Once the master dataset is established, NLP preprocessing and exploratory text analysis can be performed. These stages form the foundation for future analytical components such as classification, sentiment analysis, urgency prediction, and topic analysis. Database and dashboard components can then be connected to the processed data.

Project planning also involves defining deliverables for different stages. Examples include source-specific normalized datasets, duplicate-analysis results, the master dataset, NLP-processed data, analytical results, system architecture, and final documentation.

A structured plan helps the team identify dependencies between tasks. For example, reliable NLP analysis depends on the availability of a validated dataset, while dashboard development depends on a stable data structure. Therefore, planning CIVIC-VOICE as a sequence of connected stages reduces development risk and provides a clear pathway from raw citizen feedback to a complete analytical platform.

4.2 Work Breakdown Structure

The Work Breakdown Structure (WBS) divides CIVIC-VOICE into smaller and manageable activities. This helps the project team understand the work required and establish relationships between individual tasks.

The first major work package is project definition, which includes problem identification, objectives, scope definition, and literature review. The second package focuses on data acquisition and includes collecting CPGRAMS, Bengaluru local complaints, survey, Twitter, and email datasets.

The third package covers data engineering. This includes source-specific ingestion, normalization, cleaning, duplicate analysis, merging, and master dataset creation. The fourth package focuses on NLP processing, including text cleaning, validation, exploratory analysis, and feature extraction.

The fifth package involves system analysis and design. Activities include requirement analysis, functional and non-functional requirements, system architecture, data-flow modelling, and database design. The sixth package focuses on implementation, including development scripts, database components, and dashboard development.

The seventh package covers testing and evaluation. This includes dataset validation, preprocessing validation, analytical evaluation, and system performance testing. The final package involves documentation, presentation preparation, ethical considerations, limitations, and future scope.

The WBS allows the team to track progress systematically and identify dependencies between activities. It also makes it easier to assign responsibilities among team members and monitor whether planned project deliverables have been completed.

4.3 Timeline

The CIVIC-VOICE project is developed through a sequence of stages, with each stage depending on the successful completion of earlier activities. The initial stage focuses on understanding the problem domain and reviewing existing citizen-feedback systems. This establishes the research background and identifies the requirements for the proposed system.

The next stage involves data collection and inspection. During this phase, the project team prepares the five feedback sources and examines their structures, fields, formats, and data-quality characteristics. Source-specific normalization and cleaning are then performed.

Following data preparation, duplicate analysis and dataset merging are carried out. The resulting master dataset is validated to identify missing values and structural inconsistencies. This stage produces the central dataset used by the subsequent NLP workflow.

The next phase focuses on text preprocessing and exploratory analysis. The raw citizen feedback is converted into cleaned text, followed by validation and feature-extraction activities such as TF-IDF. These results provide evidence for the quality and usefulness of the processed data.

The later stages involve system design and implementation, including database and dashboard components where applicable. Testing and evaluation are then performed before final documentation.

The timeline can be represented using a Gantt chart showing activities, durations, dependencies, milestones, and deliverables. This provides a visual representation of project progress and helps ensure that development remains aligned with the planned objectives.

4.4 Team Responsibilities

Team responsibilities in CIVIC-VOICE are organized according to the different technical and documentation requirements of the project. Since the system combines data processing, NLP, database development, visualization, and report preparation, dividing responsibilities allows the team to work on different components in parallel while maintaining integration between them.

Data-related responsibilities include collecting the datasets, inspecting their structures, developing source-specific normalization scripts, performing cleaning, and analysing duplicate records. These activities are particularly important because the quality of the final system depends on the reliability of the underlying data.

NLP-related responsibilities include text preprocessing, validation, exploratory analysis, feature extraction, and preparation for future classification or sentiment-analysis models. These tasks require consistency in preprocessing so that the resulting features can be compared across different feedback sources.

System-development responsibilities include designing the architecture, defining database structures, developing the dashboard, and connecting analytical outputs to user-facing components. Documentation responsibilities include maintaining the methodology, recording experimental results, preparing figures and tables, and ensuring that the final report accurately reflects the implemented work.

The exact allocation of responsibilities can be presented in a team-responsibility table containing team-member names, assigned modules, major tasks, and deliverables. Clear responsibility allocation improves coordination and helps the team identify ownership of each project component.

5. ANALYSIS AND DESIGN

5.1 Requirement Analysis

Requirement analysis identifies the capabilities required for CIVIC-VOICE to process and analyse citizen feedback effectively. The requirements are derived from the central problem of fragmented feedback and the need to transform heterogeneous information into a unified analytical resource.

The system requires the ability to accept data from multiple feedback sources. Since the current project uses CPGRAMS, Bengaluru local complaints, survey responses, Twitter, and email data, the system must accommodate different source structures. It must also provide mechanisms for source-specific normalization and common-field mapping.

Data-quality requirements include the ability to detect missing values, examine duplicate records, validate fields, and verify the integrity of the merged dataset. The system must preserve source information so that feedback can later be analysed according to its origin.

NLP requirements include the ability to process textual feedback, generate cleaned text, validate the preprocessing results, and support feature extraction. The system should also provide a foundation for future analytical functions such as categorization, sentiment analysis, urgency prediction, and topic analysis.

From a user perspective, the future system should provide meaningful summaries and visualizations of citizen feedback. Performance, reliability, security, privacy, and maintainability are also important non-functional considerations.

Requirement analysis therefore ensures that the architecture and implementation are designed around the actual objectives of CIVIC-VOICE rather than focusing only on individual scripts or datasets.

5.2 Functional Requirements

Functional requirements describe the operations that CIVIC-VOICE should perform. The first functional requirement is multi-source data ingestion, allowing the system to process feedback obtained from different channels. The system should identify the source associated with each record and preserve relevant information during processing.

The second requirement is data normalization. The system should convert source-specific structures into a common representation suitable for integration. It should also perform cleaning operations to improve consistency and prepare the information for analysis.

The third requirement is duplicate analysis. The system should identify repeated records or potential duplication patterns and provide information that can be used when constructing the master dataset.

Another requirement is dataset merging. After individual datasets are processed, the system should combine them into a unified master dataset. The current CivicVoice workflow produces a master dataset containing approximately 205,168 records.

The system should also perform validation. It should examine record counts, missing values, text availability, and other relevant quality indicators. For textual feedback, the system should provide NLP preprocessing functionality and generate cleaned text.

Analytical functionality should support exploratory text analysis and feature extraction such as TF-IDF. Future functional requirements may include sentiment classification, complaint categorization, urgency prediction, topic modelling, and dashboard-based filtering.

These requirements define the core operations of the system and provide a basis for testing whether each component performs its intended function.

5.3 Non-Functional Requirements

Non-functional requirements describe the quality characteristics that CIVIC-VOICE should maintain in addition to its functional capabilities. One important requirement is reliability. The data-processing pipeline should produce consistent results when the same input data and processing steps are used.

Scalability is another important requirement because citizen-feedback datasets can grow significantly over time. The system should therefore be designed so that additional records and new feedback sources can be incorporated without requiring a complete redesign of the processing pipeline.

Maintainability is supported through modular processing scripts. Source-specific operations are separated into individual modules, allowing changes to one dataset-processing procedure without unnecessarily affecting other components.

Performance is also relevant. Data-processing operations should be capable of handling large datasets efficiently, while future dashboard components should provide responses within reasonable time for common queries and filters.

Security and privacy are particularly important because citizen feedback may contain personal or sensitive information. The system should use appropriate access controls, secure storage, and responsible handling of personally identifiable information.

Usability is another requirement for the future dashboard. Information should be presented using understandable charts, filters, summaries, and labels.

Finally, reproducibility is important for an academic project. Data transformations, preprocessing operations, validation procedures, and analytical methods should be documented so that the results can be reproduced and verified.

5.4 System Architecture

The CIVIC-VOICE system architecture represents the flow of information from multiple citizen-feedback sources to analytical outputs. The architecture is organized into several logical layers.

The first layer is the data-source layer, which contains CPGRAMS, Bengaluru local complaints, survey data, Twitter feedback, and email feedback. These sources provide different types of citizen information and may use different formats and schemas.

The second layer is the data-ingestion and normalization layer. Source-specific scripts process the individual datasets and convert their fields into a compatible representation. Cleaning and duplicate analysis are performed at this stage.

The third layer is the integration layer, where the normalized datasets are merged into the master CivicVoice dataset. Validation is then performed to confirm the quality and consistency of the integrated data.

The fourth layer is the NLP and analytics layer. Text preprocessing produces cleaned feedback, which can then be analysed using frequency analysis, TF-IDF, and future NLP models. Additional analytical functions can include categorization, sentiment analysis, and urgency prediction.
The fifth layer is the storage and presentation layer. A database such as MySQL can be used to store structured processed data, while a Streamlit-based dashboard can provide interactive visualization and filtering.

This layered architecture separates data collection, processing, analytics, storage, and presentation, making the system easier to maintain and extend.

5.5 Data Flow Diagram

The Data Flow Diagram (DFD) illustrates how citizen feedback moves through the CIVIC-VOICE system. At the highest level, the external data sources provide information to the system. These include government grievance data, local complaints, surveys, Twitter, and email.

The first process is data ingestion, where the source data is loaded into the processing environment. The next process is source-specific normalization, which converts different structures into a common representation. Cleaning operations are then applied to address inconsistencies and prepare the data for integration.

The processed datasets are passed to the duplicate-analysis stage. Potential duplicate records are identified and examined before the data is merged. The resulting datasets are combined into the master dataset.

The master dataset is then passed to the validation process. Validation examines record counts, missing values, text fields, and other quality characteristics. Once validated, the text data is passed through the NLP preprocessing stage.

The cleaned text can then be used for exploratory analysis and feature extraction such as TF-IDF. Analytical outputs can subsequently be stored in a database and presented through a dashboard.

The DFD can therefore be represented as:

Data Sources → Ingestion → Normalization → Cleaning → Duplicate Analysis → Merging → Master Dataset → Validation → NLP Processing → Analysis → Database/Dashboard.

This flow provides a clear representation of the complete CivicVoice processing pipeline.

5.6 Database Design

Database design defines how the processed CivicVoice information can be stored and accessed efficiently. Although the current project focuses strongly on dataset preparation and NLP processing, a structured database provides an important foundation for future deployment and dashboard integration.

The central database can contain a master feedback table storing normalized citizen-feedback records. Important fields may include a unique record identifier, feedback text, cleaned text, source, timestamp, location, urgency, and other attributes available in the master dataset.

Source information should be retained so that users can perform source-wise analysis. Similarly, the original text and processed text can be maintained separately where appropriate. This allows the original feedback to remain available for reference while the cleaned representation is used for NLP analysis.

Additional tables can be introduced for categories, analytical results, or model predictions as the system develops. Relationships between these tables can reduce unnecessary duplication and make future extensions easier.

Database design should also consider indexing fields that are frequently used for filtering, such as source, date, location, and category. Access controls should be applied to protect citizen information.

The database therefore acts as the storage layer between the data-processing pipeline and future analytical interfaces. A well-structured schema allows the 205,168-record master dataset and subsequent analytical outputs to be accessed efficiently while maintaining data consistency and supporting future dashboard development.

6. IMPLEMENTATION

6.1 Development Environment

The CIVIC-VOICE development environment consists of software tools and programming technologies used for data processing, NLP, analysis, and future system integration. Python forms the primary programming environment because it provides libraries and tools suitable for handling structured datasets, text processing, statistical analysis, and machine learning.

The project is organized into modular scripts so that different processing tasks can be executed independently. Scripts are used for source-specific normalization, dataset merging, duplicate analysis, validation, NLP preprocessing, and text exploration.

Data files are maintained separately from processing scripts to preserve a clear distinction between raw, processed, and master datasets. This organization improves reproducibility and makes it easier to trace how the final dataset was produced.

Python-based data-processing libraries can be used for operations such as reading datasets, manipulating tables, calculating statistics, and preparing text. NLP libraries can support tokenization and feature extraction where required.
The project also considers MySQL as the database layer and Streamlit as the dashboard technology. These components provide a potential path from the current data-processing workflow toward a deployable citizen-feedback analysis platform.

The development environment should be documented with the programming language version, major libraries, database software, dashboard framework, operating environment, and hardware configuration used during development. Maintaining this information improves reproducibility and allows other developers or evaluators to understand the technical environment in which the CivicVoice system was developed.

6.2 Dataset Modules

The dataset modules are responsible for processing individual feedback sources before they are integrated into the CivicVoice master dataset. Since each source has different characteristics, separate modules provide greater control over source-specific transformations.

The current project contains processing components for CPGRAMS, Bengaluru local complaints, survey data, Twitter data, and email feedback. Each module is responsible for reading its respective source, identifying relevant fields, applying appropriate normalization, and preparing the resulting data for integration.

For example, the CPGRAMS module handles the structure of grievance data, while the local-complaint module processes the fields associated with Bengaluru feedback. Survey data may contain more structured responses, whereas Twitter and email data can contain comparatively free-form text.
The modular design prevents source-specific logic from becoming mixed together in one large processing script. It also allows individual datasets to be updated or reprocessed independently.

After processing, each module produces a normalized representation compatible with the common CivicVoice schema. Source identifiers are retained so that records can be traced back to their origin.
The dataset modules therefore form the first major implementation layer of the project. They transform raw heterogeneous data into consistent intermediate datasets that can subsequently undergo duplicate analysis, merging, validation, and NLP processing.

6.3 Data Processing Modules

The data-processing modules implement the core data-engineering workflow of CIVIC-VOICE. These modules include normalization, duplicate analysis, merging, master-dataset validation, and related data-quality operations.

Source-specific normalization modules first convert the individual datasets into compatible formats. The duplicate-analysis module then examines repeated records and provides information about potential redundancy. The merge module combines the processed datasets into a unified dataset.

After merging, the validation module examines the resulting master dataset. Important checks include total record count, column structure, missing values, and availability of important fields. These checks help identify problems before the data is passed to NLP analysis.

The current implementation produces a master dataset containing approximately 205,168 records across seven columns. Validation indicates that the dataset contains substantial missing information in some fields, particularly timestamp, location, and urgency-related attributes. These results are documented as part of the data-quality assessment.

The modular structure also improves maintainability. If a new feedback source is introduced, a new normalization module can be developed without rewriting the entire pipeline. Similarly, changes to the duplicate-analysis procedure can be made independently of the text-processing module.

The data-processing modules therefore provide the engineering backbone of CIVIC-VOICE and ensure that heterogeneous feedback is converted into a structured dataset before NLP and analytical operations are performed.

6.4 NLP Module

The NLP module is responsible for preparing citizen-feedback text for computational analysis. The module operates on the master dataset after the data-engineering stages have been completed.

The main function of the current NLP module is text preprocessing. Raw feedback can contain variations in capitalization, punctuation, spacing, special characters, and other textual characteristics. The preprocessing workflow applies the required cleaning operations to create a standardized text representation.

The current implementation processes all 205,168 records in the master dataset. The same number of records is retained in the output, demonstrating that the preprocessing workflow does not remove records as part of the basic cleaning process. Approximately 184,072 records were modified during preprocessing, while only one record resulted in empty cleaned text.

The NLP module also supports validation by comparing the original and processed text. Text-length statistics are examined to determine how preprocessing changes the textual representation.

The cleaned text can subsequently be passed to feature-extraction methods such as TF-IDF. The module also provides the foundation for future NLP capabilities, including sentiment analysis, complaint categorization, urgency prediction, topic modelling, and keyword extraction.

The NLP module is therefore an important bridge between the structured data-engineering pipeline and the higher-level intelligence features planned for CIVIC-VOICE.

6.5 Master Dataset Generation

Master dataset generation is the central integration stage of the CivicVoice implementation. It combines the processed outputs of the individual source modules into a single dataset that can be used for common analysis.

The process begins with the normalized CPGRAMS, Bengaluru local complaints, survey, Twitter, and email datasets. These datasets are aligned with the common structure and then merged. Source identifiers are retained to distinguish records originating from different channels.

Following the merge, validation is performed to check the resulting structure. The final master dataset contains approximately 205,168 records and seven columns. This represents the combined citizen-feedback information used in the current analysis.

The generated dataset is subsequently used as the input for the NLP preprocessing stage. Maintaining the master dataset separately from the raw and intermediate datasets makes the overall workflow easier to reproduce and audit.

The master dataset also provides a single point of reference for statistical analysis. Source-wise distributions, missing-value patterns, urgency information, text lengths, and other characteristics can be examined consistently.

The generation process therefore represents one of the major outputs of the current project. Rather than analysing each feedback source independently, CIVIC-VOICE establishes a common dataset that allows multiple sources to be processed using shared analytical procedures. This dataset can later become the primary source for database storage, dashboard development, and machine-learning models.

6.6 Database Implementation

Database implementation provides the storage infrastructure required for transforming the CivicVoice data pipeline into a persistent application. MySQL is considered as the relational database layer for storing structured feedback and analytical information.

The central database can contain a feedback table corresponding to the master dataset. Fields can include a unique identifier, source, feedback text, cleaned text, timestamp, location, urgency, and other relevant attributes. Additional tables can later store categories, model predictions, analytical summaries, or user-defined metadata.

The database implementation should maintain relationships between these components while avoiding unnecessary duplication. Appropriate indexing can improve performance when users search or filter large numbers of feedback records.

Data validation rules can also be incorporated into the database layer to prevent invalid records from being inserted. Access control is important because citizen feedback may contain personally identifiable or sensitive information.At the current project stage, the database should be clearly distinguished from the completed dataset-processing work. If the MySQL component is still under development, it should be presented as an implementation stage rather than as a fully deployed system.

Once implemented, the database can provide a persistent storage layer for the 205,168-record master dataset and can serve as the backend for the Streamlit dashboard. This would allow the analytical pipeline to move from file-based processing toward a more integrated application architecture.

6.7 Dashboard Implementation

The dashboard is intended to provide a user-friendly interface for exploring and understanding citizen-feedback data. Streamlit is considered as the visualization framework because it can connect Python-based analysis with interactive web interfaces.

A CivicVoice dashboard can provide summary statistics and visualizations based on the master dataset. Possible components include total feedback records, source-wise distributions, urgency distributions, location-based summaries, frequently occurring terms, and other analytical indicators.

Interactive filters can allow users to select a feedback source, time period, location, or other available attributes. The dashboard can also provide access to cleaned feedback and NLP-based analytical results.
The dashboard should be designed to present information clearly rather than simply displaying raw data. Charts, tables, summary cards, and search functions can help users identify patterns in large datasets.

If the dashboard is not yet fully implemented, the report should present this section as the current development or planned implementation stage. The existing data-processing pipeline already provides the required foundation because the master dataset and NLP-processed data are available.

Future versions can connect the dashboard to MySQL so that users can query the database directly. Additional features such as complaint categorization, sentiment analysis, urgency prediction, and trend monitoring can also be integrated after the corresponding models have been developed and evaluated.

7. RESULTS AND EVALUATION

7.1 Dataset Statistics

The current CIVIC-VOICE implementation combines feedback from five major sources: CPGRAMS, Bengaluru local complaints, survey responses, Twitter, and email. The individual datasets vary considerably in size and structure, reflecting the different ways in which citizens communicate with public-service organizations.

After source-specific processing and merging, the final master dataset contains approximately 205,168 records and seven columns. The dataset provides a common structure for subsequent analysis and contains information collected from different feedback channels.

The individual source contributions include approximately 175,784 CPGRAMS records, 14,517 Bengaluru local complaint records, 10,000 survey records, 3,672 Twitter records, and 1,200 email records. These source-wise statistics demonstrate the different scales represented within the integrated dataset.

The master dataset also contains different levels of data completeness. Some attributes are available for most records, while others are missing for substantial portions of the data. This is particularly important when interpreting fields such as timestamp, location, and urgency.

Dataset statistics therefore provide an initial understanding of the scale and characteristics of the CivicVoice data. They also establish a baseline for subsequent data-quality, preprocessing, and analytical evaluation.

These results demonstrate that the project has successfully moved beyond individual source files toward a unified multi-source citizen-feedback dataset suitable for NLP preprocessing and future analytical modelling.

7.2 Data Quality Results

Data-quality evaluation examines the completeness and consistency of the CivicVoice master dataset. The current validation identified several important missing-value patterns that should be documented as part of the project results.

The master dataset contains approximately 205,168 records. Text availability is generally strong, allowing the dataset to be used for NLP preprocessing. However, some metadata fields contain substantial missing information. Approximately 175,779 timestamp values are unavailable, approximately 56,646 location values are unavailable, and approximately 180,651 urgency values are unavailable.

The urgency field contains approximately 10,757 records labelled Low, 6,950 labelled Medium, 4,895 labelled High, and 1,915 labelled Critical, while the remaining records do not have an urgency value. This distribution indicates that urgency cannot be assumed to be available across all feedback sources.

These missing values do not necessarily represent processing errors. In many cases, they reflect the information available in the original sources. Therefore, the evaluation distinguishes between missing information and invalid information.

The results demonstrate the importance of documenting data quality before applying predictive models. Future modelling work may require suitable strategies for handling missing attributes or may focus only on records containing the required information.

Overall, the data-quality evaluation provides transparency about the limitations and characteristics of the master dataset and establishes an evidence-based foundation for subsequent analysis.

7.3 Duplicate Analysis Results

Duplicate analysis was performed to examine repeated records within the source datasets before final integration. This evaluation is important because repeated records can influence frequency-based statistics and may cause certain issues to appear more frequently than they actually occur.

The analysis uses dedicated processing procedures to identify duplicate patterns within the available datasets. The results are considered during the construction of the master dataset. Duplicate analysis is performed before or alongside merging so that repeated records can be investigated at the source level.

The interpretation of duplicates requires care. Two records containing similar or identical text are not necessarily duplicates because citizens may independently report the same public-service issue. Conversely, repeated records with identical identifying information may indicate an actual duplicate entry.

Therefore, duplicate detection is treated as a data-quality analysis rather than simply deleting every similar record. The available identifiers, source information, text, timestamps, and other fields should be considered when determining whether records represent repeated data.

The results of duplicate analysis can be presented using source-wise counts of duplicate and unique records. Such a table would show how much duplication exists within each input dataset and how the issue was handled.

This analysis improves confidence in the master dataset and provides important documentation for later NLP and statistical analysis. It also demonstrates that the CivicVoice pipeline considers data quality before generating analytical conclusions.

7.4 NLP Preprocessing Results

The NLP preprocessing evaluation demonstrates the effect of the text-cleaning pipeline on the CivicVoice master dataset. The preprocessing stage was applied to all 205,168 records, and the resulting dataset retained the same number of records.

The analysis found that approximately 184,072 records had changes applied to their textual representation. This indicates that most of the feedback required some form of text normalization or cleaning before further NLP analysis. Only one record resulted in empty cleaned text.

Text-length statistics also provide evidence about the effect of preprocessing. The original feedback text has an average length of approximately 729.10 characters, with a minimum of 1 character and a maximum of approximately 7,252 characters. After preprocessing, the average text length is approximately 715.68 characters. The reduction in average length indicates that unnecessary textual elements were removed or standardized while retaining the majority of the textual information. Importantly, the preprocessing stage did not reduce the overall record count.

These results indicate that the NLP preprocessing workflow is suitable for preparing the master dataset for feature extraction and exploratory text analysis. The validation results also provide measurable evidence that the preprocessing pipeline has operated consistently.
Further evaluation can examine token counts, vocabulary size, stop-word distribution, and other linguistic characteristics before and after preprocessing.

7.5 Text Analysis Results

Text analysis provides an initial understanding of the language contained in citizen feedback. After preprocessing, the cleaned text can be examined to identify frequently occurring words, common expressions, text lengths, and patterns across different feedback sources.

The current dataset contains substantial variation in feedback length. Original text ranges from very short entries to long citizen descriptions, with an average length of approximately 729 characters. After preprocessing, the average decreases to approximately 716 characters.

Frequency-based analysis can identify terms that appear repeatedly throughout the dataset. However, frequently occurring words are not necessarily the most informative because common terms may appear across a large proportion of documents. For this reason, simple frequency analysis can be complemented with TF-IDF analysis.

Text analysis can also be performed separately for different feedback sources. This can reveal differences in vocabulary between formal grievance submissions, local complaints, surveys, social-media feedback, and emails.

The current text analysis primarily establishes an exploratory understanding of the dataset. It should not be interpreted as a complete semantic analysis of citizen concerns. More advanced techniques such as topic modelling, sentiment analysis, named-entity recognition, and classification can be introduced in later stages.

The results from this stage help verify that the cleaned dataset contains meaningful textual information and provide a basis for selecting appropriate NLP methods for subsequent CivicVoice development.

7.6 TF-IDF Results

TF-IDF analysis is used to identify terms that are relatively informative within the collection of citizen-feedback documents. Unlike simple frequency counts, TF-IDF considers both how frequently a term appears within a document and how widely it appears across the entire dataset.

The TF-IDF representation converts the cleaned citizen feedback into numerical features. Each document is represented using a vector of term weights. Terms that appear frequently in one document but less frequently across the overall collection can receive relatively higher weights.

Within CIVIC-VOICE, this technique provides an analytical method for identifying important vocabulary in the large master dataset. The resulting terms can be examined overall or separately for different feedback sources, allowing patterns to be compared across channels.
TF-IDF can also serve as a feature representation for future machine-learning tasks. For example, classification models can use the resulting feature matrix as input when sufficient labelled training data is available.

The TF-IDF results should be presented using a table or visualization showing the most informative terms and their corresponding weights. Interpretation should consider the preprocessing choices, vocabulary restrictions, and source distribution.
TF-IDF therefore represents an important connection between the current exploratory NLP work and future predictive modelling. It transforms the cleaned citizen feedback into a structured numerical representation while retaining information about the relative importance of terms across documents.

7.7 Model Evaluation

Model evaluation is required when CIVIC-VOICE is extended with machine-learning or NLP prediction models. Possible future models may perform tasks such as sentiment classification, complaint categorization, urgency prediction, or topic classification.

A reliable evaluation requires an appropriate labelled dataset. The current master dataset contains urgency information for only a subset of records, while many records do not contain an urgency value. Therefore, model evaluation should not treat missing values as labelled examples.

For a classification task, the dataset should be divided into training, validation, and testing subsets where appropriate. Evaluation metrics can include accuracy, precision, recall, and F1-score. A confusion matrix can also be used to identify which classes are being confused by the model.

For imbalanced classes, accuracy alone may not provide a complete picture of model performance. Class-wise precision, recall, and F1-score should therefore be examined.

At the current stage, the report should distinguish between the completed data-processing and NLP-preprocessing results and future model evaluation. Numerical model-performance values should only be reported after an actual model has been trained and evaluated on a defined test dataset.

This approach ensures that the final report does not present unverified model performance and provides a clear pathway for evaluating future CivicVoice intelligence components.

7.8 System Performance

System performance evaluation examines how efficiently the CivicVoice processing pipeline handles large-scale citizen-feedback data. Performance considerations are particularly important because the current master dataset contains approximately 205,168 records.

The processing pipeline includes several stages, including source ingestion, normalization, duplicate analysis, merging, validation, and NLP preprocessing. Each stage may have different computational requirements. Performance can therefore be evaluated by measuring execution time, memory usage, and output consistency.

For the NLP preprocessing stage, the project successfully processes all 205,168 records while retaining the same number of output records. This demonstrates that the current preprocessing workflow can operate on the full master dataset without reducing the dataset size.

Future system-level evaluation can measure the execution time required for complete dataset processing and the time required to perform common analytical operations. Database performance can similarly be evaluated after MySQL integration by measuring query response time for filtering and aggregation operations.

Dashboard performance can be assessed using page-load time and response time for common filters and visualizations.System performance should therefore be evaluated using measurable indicators rather than subjective descriptions. These results can help identify processing bottlenecks and guide future optimization as additional data sources and analytical models are incorporated into CIVIC-VOICE.

8. SOCIAL, LEGAL, ETHICAL, SUSTAINABILITY AND SAFETY

8.1 Social Impact

CIVIC-VOICE has potential social value because it provides a structured method for analysing citizen feedback collected through multiple communication channels. Citizen feedback can contain information about problems experienced with public services, and organizing this information can make it easier to identify recurring themes and areas requiring attention.

The project focuses on combining information from government grievance systems, local complaints, surveys, social media, and email. This multi-source approach recognizes that citizens communicate through different channels and that no single source necessarily represents the complete range of feedback.

The system can support public-service analysis by providing a common dataset from which trends, frequently discussed topics, and source-wise patterns can be examined. Future dashboard functionality can make these findings easier to explore.

However, the system should not be treated as a replacement for human decision-making. Automated analysis can contain errors, particularly when dealing with informal language, multilingual content, sarcasm, incomplete information, or ambiguous complaints. Therefore, analytical results should support human review rather than automatically determine public-service decisions.

The social impact of CIVIC-VOICE depends on responsible implementation. Appropriate privacy safeguards, transparent analytical methods, and careful interpretation are required to ensure that citizen feedback is used constructively. The project can therefore contribute to improved understanding of citizen concerns while maintaining appropriate safeguards around the use of citizen-generated information.

8.2 Privacy

Privacy is an important consideration for CIVIC-VOICE because citizen feedback may contain personal information. Email messages, social-media posts, complaint descriptions, and survey responses may include names, contact information, addresses, usernames, or other information that can identify individuals.

The system should therefore follow a data-minimization approach. Only information required for the intended analysis should be retained, and unnecessary personally identifiable information should be removed or anonymized where possible.

During preprocessing, personal information should not be exposed unnecessarily in analytical outputs or dashboard visualizations. User-facing interfaces should focus on aggregated information rather than displaying sensitive individual records unless there is a legitimate and authorized reason to do so.
Access controls should be implemented for stored data, particularly if the system is connected to a database. Sensitive raw data should have stronger access restrictions than aggregated analytical results.

Privacy considerations are also important when using social-media information. Public availability of information does not automatically mean that every piece of information should be reproduced or redistributed without consideration of context and applicable rules.

CIVIC-VOICE should therefore treat privacy as a design requirement rather than an afterthought. Data collection, storage, preprocessing, database access, and dashboard presentation should all be designed to reduce unnecessary exposure of personal information while retaining the information required for legitimate citizen-feedback analysis.

8.3 Data Protection

Data protection involves safeguarding citizen-feedback information against unauthorized access, accidental loss, inappropriate modification, and unnecessary disclosure. Because CIVIC-VOICE combines data from multiple sources, the protection of the integrated dataset is particularly important.

The system should maintain appropriate separation between raw data, processed data, and analytical outputs. Raw datasets may contain more identifying information than is required for analysis and should therefore receive appropriate access restrictions.

If MySQL is used as the storage layer, authentication and authorization should be implemented so that only authorized users can access sensitive information. Database backups should also be protected, and data transfers should use secure mechanisms where applicable.

Data integrity is another component of protection. Validation procedures should help ensure that records are not unintentionally altered or corrupted during ingestion, normalization, merging, and preprocessing. Maintaining processing logs and reproducible scripts can help identify how data was transformed.

The project should also define retention and deletion procedures where applicable. Data should not be retained indefinitely without a legitimate purpose, particularly when it contains personal information.

CIVIC-VOICE can therefore incorporate data protection through access control, secure storage, controlled processing, data minimization, validation, and appropriate retention practices. These measures help ensure that the technical benefits of multi-source citizen-feedback analysis do not create unnecessary privacy or security risks.

8.4 Ethical Considerations

Ethical considerations are important because CIVIC-VOICE analyses information generated by citizens and may eventually produce automated interpretations of their feedback. The system should therefore be designed to support responsible use of citizen-generated data.

One important consideration is fairness. NLP models may perform differently across languages, writing styles, demographic groups, or types of feedback. If future models are trained using biased or incomplete datasets, their predictions may not accurately represent all citizens.Transparency is another consideration. Users should understand that automated classifications, sentiment scores, or urgency predictions are analytical outputs rather than guaranteed facts. Where automated predictions are used, appropriate explanations and human review should be considered.

Another ethical issue is the handling of sensitive complaints. Citizen feedback may describe personal experiences, disputes, or potentially sensitive circumstances. Such information should not be unnecessarily exposed through dashboards or analytical reports.The project should also avoid allowing automated outputs to become the sole basis for decisions that significantly affect citizens. Future AI components should assist human analysis rather than replace appropriate human oversight.

Finally, the project should clearly document the limitations of the dataset and analytical models. Responsible interpretation requires acknowledging missing data, source differences, and possible model errors. These principles can help ensure that CIVIC-VOICE is developed as a responsible citizen-feedback analysis platform.

8.5 Legal Considerations

Legal considerations arise from the collection, processing, storage, and analysis of citizen feedback. The exact legal requirements depend on the source of the data, the jurisdiction, the type of information collected, and how the system is deployed.

CIVIC-VOICE should therefore consider applicable data-protection and privacy requirements when handling personal information. Data obtained from government systems, surveys, email, and social-media platforms may be subject to different terms, permissions, or restrictions.

The project should ensure that data is collected and used for legitimate purposes and that access to personal information is appropriately controlled. Where required, identifying information should be anonymized or minimized before analysis.

Copyright and platform terms should also be considered when using information obtained from online services. The use of social-media data should follow the applicable platform policies and any relevant permissions or restrictions.

If the system is eventually deployed for public-service decision support, additional considerations may apply regarding transparency, accountability, record keeping, and automated decision-making.The report should therefore describe legal considerations without assuming that a single rule applies to every dataset. The final implementation should be reviewed against the applicable laws, institutional policies, dataset licences, platform terms, and project requirements.

Legal compliance is consequently an ongoing responsibility throughout the CivicVoice lifecycle, from data collection and processing to storage, visualization, and eventual deployment.

8.6 Sustainability

Sustainability in CIVIC-VOICE can be considered from technical, operational, and social perspectives. From a technical perspective, the system should be designed using modular components so that new feedback sources can be added without requiring the entire system to be redesigned.

The current modular data-processing approach supports this objective. Separate normalization scripts allow individual sources to be processed independently, while the common master-dataset structure allows the resulting information to be integrated.

Sustainable development also requires efficient use of computational resources. Data-processing pipelines should avoid unnecessary repeated processing, and database queries and dashboard operations should be optimized as the dataset grows.

From a social perspective, a system that continuously organizes citizen feedback can support longer-term monitoring of public-service concerns. However, sustainability also depends on maintaining data quality and ensuring that analytical outputs remain relevant as citizen communication patterns change.The project should also consider maintainability. Documentation, version-controlled scripts, clear data schemas, and reproducible workflows make it easier for future teams to understand and extend the system.

CIVIC-VOICE can therefore be designed as an extensible platform rather than a one-time data-analysis project. Future sources, analytical models, and visualization features can be incorporated as requirements evolve. This approach supports the long-term usefulness of the project while reducing the need to rebuild the complete system whenever new requirements arise.

9. CONCLUSION

9.1 Conclusion

CIVIC-VOICE presents a structured approach for integrating and analysing citizen feedback collected from multiple channels. The project addresses the challenge of fragmented feedback by combining information from CPGRAMS, Bengaluru local complaints, survey responses, Twitter, and email into a unified processing framework.

A major outcome of the current work is the creation of a master dataset containing approximately 205,168 records across seven columns. The project applies source-specific normalization, data cleaning, duplicate analysis, merging, and validation before creating the master dataset. This process establishes a consistent foundation for subsequent analysis.

The NLP component further processes citizen-feedback text to create a cleaned representation suitable for computational analysis. The current preprocessing workflow retains all 205,168 records, with approximately 184,072 records receiving textual changes and only one record resulting in empty cleaned text. Text-length analysis also demonstrates the effect of preprocessing.

The project establishes the foundation for exploratory text analysis and TF-IDF-based feature extraction. Future extensions can incorporate automated categorization, sentiment analysis, urgency prediction, topic modelling, database integration, and interactive visualization.

Overall, the current work demonstrates how data engineering and NLP preprocessing can be combined to create a scalable foundation for multi-channel citizen-feedback analysis. The resulting pipeline and master dataset provide the basis for developing the complete CivicVoice analytical platform.

9.2 Limitations

Although CIVIC-VOICE provides a structured framework for multi-source citizen-feedback analysis, several limitations exist in the current implementation. The first limitation is variation in data quality between sources. The five datasets have different structures, levels of completeness, and types of information. Consequently, some fields cannot be consistently populated across all records.

The master dataset contains substantial missing information. Approximately 175,779 timestamp values, 56,646 location values, and 180,651 urgency values are unavailable. These missing values reflect limitations in the source data and may restrict certain types of analysis.

Another limitation is the heterogeneous nature of citizen language. Feedback may contain informal expressions, abbreviations, spelling variations, multilingual content, and context-dependent meanings. Basic preprocessing cannot completely resolve all such linguistic complexities.

The current work also establishes the foundation for advanced NLP analysis rather than providing fully evaluated predictive models for every proposed feature. Sentiment classification, automated categorization, urgency prediction, and topic modelling require suitable labelled datasets and rigorous evaluation.

Similarly, database and dashboard components may require further development before full deployment.

These limitations do not invalidate the current results; instead, they define the boundaries of the current implementation and identify areas that require additional work before CIVIC-VOICE can be considered a fully deployed intelligent citizen-feedback platform.

9.3 Future Scope

CIVIC-VOICE can be extended in several directions to develop a more comprehensive citizen-feedback intelligence platform. One important area is automated feedback categorization. NLP and machine-learning models can be developed to classify complaints into service-related categories once an appropriate labelled dataset is available.

Sentiment analysis can also be incorporated to identify positive, negative, or neutral expressions within citizen feedback. Similarly, urgency prediction can be developed to estimate the priority of complaints where reliable labelled examples are available.

Topic modelling and keyword extraction can help identify recurring themes across large collections of feedback. These techniques can support the discovery of emerging issues that may not be captured through predefined categories.Another area of future development is multilingual NLP. Since citizen feedback may be submitted in different languages, multilingual preprocessing and language-specific models could improve the system's ability to analyse diverse feedback.

The project can also be extended through MySQL database integration and an interactive Streamlit dashboard. These components can allow users to search, filter, visualize, and monitor citizen feedback more efficiently. Real-time or scheduled data ingestion from supported sources is another possible extension. Additional analytical capabilities such as trend detection, alerts, geographic analysis, and AI-assisted querying could also be introduced.

Future development should continue to prioritize privacy, security, transparency, and human oversight. These improvements can transform the current CivicVoice data-processing pipeline into a more comprehensive platform for multi-channel citizen-feedback analysis.

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List of Abbreviations (16)

Abbreviation	Full Form

10. REVIEW 4 IMPLEMENTATION UPDATE

This section updates the earlier report to reflect the work completed or initiated by the team by Review 4. Transformer-based NLP is the intended analysis approach discussed for the project; TF-IDF should not be presented as the main model used by CIVIC-VOICE. TF-IDF is a separate traditional text-feature method and is not interchangeable with a transformer model.

10.1 MySQL Database Progress

A MySQL database has been created locally and tables have been designed for storing citizen feedback and related analysis information. The database is intended to support structured retrieval and integration with the application. The final schema, table names, keys and relationships should be documented from the current SQL files rather than inferred.

10.2 FastAPI Backend Progress

FastAPI backend development has been started, and the team has run a local Uvicorn server and checked a health endpoint during setup. A responding health endpoint confirms that the API process is available, but does not alone prove that database queries or dashboard integration are complete. The next check is to confirm that endpoints retrieve actual MySQL records and return consistent responses.

10.3 Dashboard Implementation

The supplied CIVIC-VOICE dashboard screenshots show a Dashboard Overview with source, urgency and location filters; summary cards; feedback-by-source chart; urgency distribution; feedback trend; high-priority feedback explorer; and an AI & NLP Insights section. The interface is a dashboard prototype. The screenshot displays 205,168 total feedback, five sources, 6,810 high-priority items and 24 locations. These figures must be reconciled with the current database/API before being described as verified live metrics.

The NLP insight cards show sentiment, issue category, topic and urgency fields, and are labelled Prototype. These cards should be described as prototype outputs unless the team can demonstrate they are produced by the actual transformer inference pipeline.

10.4 Updated Results and Insights

The previous report documents a master dataset of approximately 205,168 records across seven columns. It reports that preprocessing preserved the record count, changed the text of approximately 184,072 records and produced one record with empty cleaned text. Average text length was reported to change from approximately 729 to 716 characters. These values are inherited from the earlier report and should be rerun against the current scripts before final submission.

The verified emphasis for Review 4 is the unified multi-source dataset, preprocessing workflow, local MySQL setup, initial FastAPI development and dashboard prototype. No transformer accuracy or other model-performance figure is asserted here because a verified evaluation result was not available in the supplied material.

10.5 Remaining Work

1. Reconcile the five source-wise counts with the master dataset count. 2. Confirm the MySQL schema and database loading. 3. Test FastAPI endpoints against actual database rows. 4. Connect dashboard filters and charts to API responses. 5. Verify that NLP cards use transformer predictions rather than placeholder values. 6. Evaluate the transformer-based tasks with a suitable labelled test set and report precision, recall, F1-score and a confusion matrix where applicable. 7. Clearly label prototype or sample data in the dashboard and report.

Corrected NLP terminology

Replace statements that present TF-IDF as the project’s primary NLP method with a description of the transformer-based approach actually pursued by the team. Keep TF-IDF only if it was used in a separate exploratory experiment and identify it as such. Do not claim a particular transformer architecture, fine-tuning procedure or accuracy unless confirmed by the project code and outputs.

Note: The existing report pages are retained to preserve the original academic layout. The Table of Contents has been revised and this Review 4 update is added as a new section. Earlier body paragraphs that mention TF-IDF are not silently rewritten in the original PDF pages; the correction above identifies how those references should be interpreted and revised in the editable source document.